

Weakly Supervised Localisation Generalises Where Genotype Prediction Does Not: An Externally Validated Null for *EGFR* Radiogenomics from Lung CT

Original Research · Manuscript type: Original Research
Word count (Introduction through Discussion): see title page
Author names and affiliations removed for anonymized review

Summary Statement

Weakly supervised tumour localisation transferred from a United States surgical cohort to a Dutch radiotherapy cohort, whereas prediction of epidermal growth factor receptor mutation status from the same lesion-anchored computed tomographic images did not exceed a five-variable model built from routinely recorded clinical data.

Key Points

- In 153 patients with a known *EGFR* call, pooled out-of-fold AUC was 0.618 ± 0.025 versus 0.794 for smoking status alone; adding the model to a five-variable clinical model changed AUC by -0.029 ± 0.015 , with every seed’s confidence interval including zero.
- Supervised attention transferred to 420 external patients from a different country and scanner fleet, with an attention-mass lift of $+0.244$ over a per-fold spatial shuffle and pointing accuracy of 0.609 versus 0.476 for a centre prior and 0.005 for an untrained head.
- Across six defensible preprocessing choices the radiomics comparator moved 0.136 in AUC, exceeding the effect sizes this literature commonly reports as findings.

Abbreviations: AUC = area under the receiver operating characteristic curve, CI = confidence interval, CV = cross-validation, IQR = interquartile range, NSCLC = non-small cell lung cancer, ROI = region of interest, TKI = tyrosine kinase inhibitor.

Abstract

Purpose. To evaluate whether a grounded vision-language model predicts *EGFR* mutation status from lung CT beyond routinely available clinical variables, and whether its supervised attention localises tumours on an unseen cohort.

Materials and Methods. This retrospective secondary analysis used the public NSCLC-Radiogenomics collection. Its CT dates are de-identification offsets, so no accrual window is recoverable; the index test-to-reference standard interval had a median of 38 days (IQR, 19.5–65.5). Of 211 subjects, 153 had a known *EGFR* call (mean age, 69 years $\pm$ 9 [SD]; 98 male, 55 female; 40 mutant [26.1%]). A BiomedCLIP ViT-B/16 encoder with a region-query grounding head supervised against tumour segmentations was trained with nested five-fold cross-validation, three seeds. The primary endpoint was external generalisation of localisation, assessed on 420 held-out NSCLC-Radiomics patients against three controls: a per-fold spatial shuffle, a centre prior, and an untrained head. A

co-primary endpoint, incremental AUC over a five-variable clinical model, was pre-declared as an expected null; the analysis plan was fixed before external evaluation. $P < .05$ indicated significance. **Results.** Pooled out-of-fold *EGFR* AUC was 0.618 ± 0.025 , versus 0.794 for smoking status alone. Adding the model to clinical variables changed AUC by -0.029 ± 0.015 , every seed’s CI including zero; no fusion topology exceeded the clinical model. Calibration slope was 0.19, and Brier score (0.236) exceeded the base-rate floor (0.193). Localisation transferred: attention-mass lift was +0.244 over the shuffle in 11 of 15 external evaluations, with pointing accuracy 0.609 versus 0.476 (centre prior) and 0.005 (untrained head).

Conclusion. Supervised localisation generalised across institutions, whereas lesion-anchored *EGFR* prediction did not exceed a five-variable clinical model.

1 Introduction

EGFR genotype directs tyrosine kinase inhibitor therapy in non-small cell lung cancer (NSCLC). Tissue is often limited and molecular testing is not universally available, so a non-invasive imaging surrogate would have real clinical value. This has motivated a large radiogenomics literature.

That literature reports *EGFR* prediction from CT at AUC 0.80–0.89 [5, 6, 8–10]. The cohorts are typically 200–826 patients, usually single-centre, and external validation is uncommon. Against this, the one large study with external validation — 1,646 patients and 11,473 segmented lesions with next-generation-sequencing labels — reports internal AUC 0.62–0.68 and external 0.55–0.63 [4]. Notably, its external cohort is NSCLC-Radiogenomics, the collection used here.

That discrepancy is this study’s motivation. Reported performance in this field is known to be sensitive to methodological choices that are invisible at internal evaluation [11, 12], which makes the separation of claims more informative than another point estimate. We therefore do not propose a better classifier. We separate two claims that are routinely bundled — that such a model *localises* the lesion, and that it *genotypes* it — and test each with the controls it requires, on a cohort where the second can be measured against the clinical variables it would have to beat.

2 Materials and Methods

2.1 Ethics

This study is a retrospective secondary analysis of publicly available, fully de-identified imaging and clinical data distributed by The Cancer Imaging Archive under open licences, and no protected health information was accessed at any point.

[AUTHOR ACTION REQUIRED BEFORE SUBMISSION] Insert the institutional review board’s written determination here — either formal approval, a waiver of informed consent, or a not-human-subjects-research / exempt determination with its reference number and date. Journal policy requires an explicit statement, and the authors’ assessment of the regulations is not a substitute for the board’s. Also state HIPAA-compliance status if any author’s institution is in the United States.

2.2 Study design and cohorts

The analysis plan, including the primary endpoint and all controls, was fixed and version-controlled before external evaluation; deviations from it are logged in the public repository accompanying this work.

The internal cohort is NSCLC-Radiogenomics [1], a surgical series distributed by The Cancer Imaging Archive. Participant flow is given in Figure 1: 211 subjects in the clinical sheet, 188 with an imaging series, 158 successfully preprocessed, 153 with a known *EGFR* call (mean age, 69 years $\pm$ 9 [SD]; 98 male, 55 female), 133 adenocarcinoma, and 97 with both adenocarcinoma histology and a pixel-level segmentation. That last set is the pre-specified primary analysis set. Cohort characteristics appear in Table 1.

No accrual window is reported. The collection’s CT dates span 1989–1996 and are de-identification offsets rather than dates; publishing a range derived from them would misdescribe the cohort. The interval from index test to reference standard, which survives the offset because it is a difference, has a median of 38 days (IQR, 19.5–65.5).

The external cohort is NSCLC-Radiomics [3], 420 preprocessed patients from a Dutch centre — a different country, scanner fleet, and clinical population (radiotherapy planning rather than surgical). The two cohorts share no patients, and no evaluated checkpoint was warm-started from the external collection; both facts are asserted in the accompanying automated test suite rather than claimed in prose.

2.3 Reference standard

EGFR status was determined by sequencing and is distributed with the collection [1]. The grounding reference standard is the expert tumour segmentation distributed with each collection. Only one contour per lesion exists, so inter-rater agreement cannot be computed; a mask perturbation sweep is reported instead and is explicitly not a substitute (Discussion).

2.4 Model

A BiomedCLIP ViT-B/16 encoder (last two blocks unfrozen) produces patch tokens per section, aggregated across 16 sections by a masked mean. A grounding head applies eight learned region queries by cross-attention, and the resulting attention maps are supervised against the tumour region of interest (ROI) projected onto the patch grid. The classifier pools from the grounded region embeddings, so prediction and localisation share a representation.

Inputs are axial crops of $(\text{bbox}_{\text{mm}} + 24) \times 2.5$ around the lesion with 30% positional jitter, windowed at centre -600 HU / width 1500 HU and resized to 224×224 .

2.5 Endpoints and controls

H₁ (primary). External attention-mass lift over a per-fold spatial shuffle is greater than zero. Three controls are mandatory in every reported table:

- **Spatial shuffle** — the same attention values permuted across cells, isolating *where* mass fell from *how peaked* the map is.
- **Centre prior** — external ROI centroids sit at (0.50, 0.51) of the frame (SD $\approx$ 0.08) covering 4.9% of area, so “look at the middle” is a strong baseline. Pointing accuracy and area-matched intersection over union are invariant to the prior’s width, so that margin cannot be tuned.
- **Weight randomisation** — an untrained head of identical architecture, ruling out architectural inductive bias.

H₂ (co-primary, pre-declared null). $\Delta\text{AUC} = \text{AUC}(\text{clinical} + \text{model}) - \text{AUC}(\text{clinical})$.

2.6 Statistical analysis

Evaluation used nested five-fold cross-validation (CV) stratified on *EGFR*, three seeds, with within-fold rank normalisation before pooling. The headline is the across-seed mean $\pm$ SD; a pooled confidence interval (CI) absorbs no rerun variance and is never quoted alone. **No operating point is defined:** the index test is continuous and the calibration reported in Section 3.2 does not support a threshold, so sensitivity and specificity are not reported.

Comparisons used a paired bootstrap over patients (2000 resamples), with Holm correction across the ten arms of the pooling sweep. Precision was computed by the Hanley–McNeil closed form [13], validated against a bootstrap; minimum sample size follows Riley et al. [14], applied only where its inputs are defined. Reporting follows the Checklist for Artificial Intelligence in Medical Imaging [15]; the completed checklist is submitted with this manuscript.

Two-sided $P < .05$ was considered to indicate a significant difference. Analyses were performed in Python 3.13.9 (Python Software Foundation) with NumPy 2.1.3, SciPy 1.15.3, scikit-learn 1.6.1, pandas 2.2.3 and statsmodels 0.14.4 (all open-source, respective maintainers), and PyTorch 2.10.0 (Meta Platforms).

3 Results

3.1 The model does not exceed the clinical variables

Table 2 gives every arm on identical patients and folds. Pooled out-of-fold *EGFR* AUC was 0.618 ± 0.025 across three seeds. Smoking status alone reached 0.794; the five-variable clinical model 0.764.

The co-primary endpoint was $\Delta\text{AUC} = -0.029 \pm 0.015$, with every seed’s CI including zero. No fusion topology exceeded the clinical model: soft voting reached 0.748, cross-fitted stacking 0.735, hard voting 0.690. Adding radiomics significantly *harmed* the combination (-0.130 , $P = .02$). A logistic regression on frozen encoder features scored 0.617 ± 0.052 — indistinguishable from the full network to three decimals.

Results are shown per arm in Figure 2 and with paired deltas in Figure 3. *KRAS* was at chance (0.489 ± 0.005); Gevaert et al. likewise report no significant *KRAS* model on this cohort [2].

3.2 The predictions are also uncalibrated

Calibration slope averaged 0.19 against a perfect 1.0, indicating predictions roughly five times too extreme. At a base rate of 26.1%, a model predicting the prevalence for every patient achieves a Brier score of 0.193; the model scored 0.236. **The probabilities are therefore worse than a constant**, with resolution — the component only a better model can improve — between 0.009 and 0.017. This is why no operating point is pre-specified, and why any future externally validated artifact must carry explicit recalibration.

3.3 On the subset that should carry it, performance is weaker

Every *EGFR*-mutant patient in this collection is an adenocarcinoma; the 17 squamous and 3 NSCLC-not-otherwise-specified patients are wild-type without exception. Including them makes 13% of the cohort classifiable from histology alone with no imaging. On the pre-specified primary analysis set — segmented adenocarcinoma, $n = 97$, 23 positives — AUC falls to 0.563. The same pattern appears in the literature: Yamazaki et al. report 0.774 across histologies but 0.687 restricted to adenocarcinoma [7].

3.4 Localisation generalises to an unseen cohort

On 420 held-out NSCLC-Radiomics patients (Figure 5), attention mass inside the tumour was 0.292 against 0.048 for the spatial shuffle — a lift of +0.244, beating the shuffle in 11 of 15 evaluations. Pointing accuracy reached 0.609 against 0.476 for the centre prior and 0.005 for a randomly initialised head, which returned a lift of exactly +0.000 at a peak-to-mean ratio of 1.01.

The margin over the pre-specified fold-rate threshold is one fold: 11 of 15 (73.3%) against a required 71.4%. Because that margin invites the reading that the endpoint is knife-edge, Table 3 reports every evaluation individually. The distribution is bimodal rather than marginal. The weakest success has a lift of +0.258 and pointing accuracy of 0.757; the strongest failure has a lift of −0.005 and pointing accuracy of exactly 0.000. No evaluation falls between 0.000 and 0.757 on pointing accuracy. Folds either localise decisively or not at all, so the endpoint is insensitive to where the threshold is placed within a wide interval, while remaining sensitive to how many folds land on each side.

The failures are also concentrated rather than scattered: one training run localised in 2 of 5 folds while the others reached 5 of 5 and 4 of 5, and that same run localises in 2 of 5 internally while carrying the *highest* classification AUC of the three. Localisation is therefore a property of most training runs rather than of the method.

Representative attention maps appear in Figure 6. These are the first six segmented patients of a held-out fold in index order, not selected on score.

3.5 The peritumoral region carries the signal, but not significantly

Because the classifier pools from the grounded region — the lesion itself — we asked whether the *EGFR* signal is located where the grounding loss supervises attention. Over ten seeds on frozen encoder features, through identical nested folds, mean-pooling the peritumoral ring scored 0.695 against 0.632 for the whole crop and 0.629 for the tumour interior. A size- and shape-matched ring at a *random* location scored 0.575, so the gain is location-specific rather than an artefact of pooling a thin annulus. Adding a peritumoral branch to the classifier moved the trained model from 0.618 to 0.675 ± 0.017 , improving in 3 of 3 seeds.

This does not survive correction, and we report it as suggestive rather than established. The sweep scores ten arms on the same folds; under a paired bootstrap over patients with Holm correction, the best arm has $P_{\text{raw}} = .03$ and $P_{\text{Holm}} = .28$. End-to-end the branch is worth +0.058 (95% CI: −0.008, +0.128; $P = .09$). The direction is consistent across frozen and trained settings and the location controls behave; the significance is absent.

It is also a replication rather than a discovery: peritumoral radiomics for *EGFR* has been reported at least six times since 2022, with optimal ring widths of 1, 2, 3, 4, 6 and 15 mm [5–10]. What is new here is the tension it exposes — the grounding objective supervises attention *onto* the lesion, and the classification signal appears to sit just outside it.

3.6 Analytic choice is worth more than the effect

On identical features and estimator, the radiomics arm moved **0.136 in AUC** across six defensible preprocessing choices, and **0.066** between a flat and a nested CV protocol. Both exceed the effect sizes this literature reports as findings, and both are of the kind that internal evaluation cannot detect [11].

3.7 More patients would not settle it

Riley’s criterion requires 414 patients for the five-variable clinical model alone and 42,362 for the 512-feature frozen probe; this cohort has 153. Neither arm is identifiable at this sample size, which is why “the probe matches the model” cannot be read as evidence that the architecture is redundant.

Precision, not power, binds any external test. At the observed prevalence a 95% CI half-width of ± 0.05 requires $n = 661$. The entire open world of *EGFR*-labelled CT holds approximately 327 cases with 64 positives, giving ± 0.154 — an interval spanning chance and the published 0.80s simultaneously.

4 Discussion

Two claims usually bundled together separate cleanly here, and separate differently. Supervised localisation transferred to a cohort from another country, scanner fleet and treatment setting. Genotype prediction did not clear a five-variable clinical model on the cohort it was trained on ($\Delta\text{AUC} = -0.029 \pm 0.015$), was not rescued by any fusion topology, and was worse than a constant on the probability scale (Brier 0.236 vs a base-rate floor of 0.193).

What the localisation endpoint does and does not establish. Attention mass, pointing accuracy and area-matched intersection over union are not measures of diagnostic accuracy, and no clinical claim follows from them directly. They answer a narrower question that we think has been assumed rather than tested: whether the explanation a model offers for its prediction is anchored to the lesion at all, on data from an institution it has never seen. That question matters precisely because the accompanying prediction fails. A saliency map that tracks the tumour on an unseen cohort while the classifier built on it performs at 0.618 is direct evidence that plausible localisation is not evidence of predictive validity — a dissociation that cannot be observed in a study where both are reported as successes. We therefore present H_1 as a methodological result about interpretability claims, not as a clinical endpoint, and we do not propose the localiser for clinical use.

Convergence with the largest external study. Rodríguez Sánchez et al. [4] is the strongest argument that this is a property of the problem rather than of our sample. Their 1,646 patients and 11,473 lesions, with next-generation-sequencing labels and biopsy-anchored training, reproduce the interval we obtain from 153 patients; their external cohort is the collection we use internally. Ten times the patients did not move the ceiling. Read together with the measured precision bound above — ± 0.154 on the largest test set the open literature can currently assemble — the implication is that the field cannot resolve this question with public data, and that the reported 0.80s are more plausibly explained by analytic variation than by a signal the larger studies missed.

Comparison with prior reports. Our peritumoral finding replicates six prior reports in direction while failing to reach significance after correction for the ten arms actually examined. We consider that a more likely description of this literature than the alternative, given that the same six reports place the optimal ring width anywhere from 1 to 15 mm. Similarly, our histology result reproduces Yamazaki et al. [7]: performance falls when the squamous patients, who are wild-type without exception, are removed. Any study reporting *EGFR* AUC across mixed histologies is partly reporting a histology classifier.

Limitations. The internal cohort is single-institution and historical, and no accrual window is recoverable from it. The classification arm has no external validation; that is justified by the measured precision bound above rather than by circumstance, and is reported as a finding rather than deferred. Inter-rater variability of the segmentations is not measured, because the source collections distribute one contour per lesion; a mask perturbation sweep (dilation, erosion, translation) is reported instead and is explicitly not a substitute for a second reader. Acquisition parameters —

section thickness, convolution kernel, vendor — are incompletely recoverable. One of three training runs largely fails to localise, and we report the endpoint on all three rather than on the two that succeed. Finally, H_1 is confirmatory of a prior observation on an independent patient set rather than a blind prospective test: the external cohort was on disk, and the metric’s behaviour on it partly known, when the threshold was set. The registered plan states this precisely, and we state it here rather than let a reader infer a stronger design than was run.

Conclusion. On this cohort, weakly supervised localisation generalised across institutions while *EGFR* prediction from lesion-anchored CT did not exceed routinely available clinical variables. Reports of AUC 0.80–0.89 should be read against the analytic-choice sensitivity and precision bounds measured here.

Data and code availability

All code, the pre-specified analysis plan with its deviation log, the completed CLAIM checklist, and every archived run result — including negative results — are available at an anonymized repository for review ([ANONYMIZED REPOSITORY LINK --- see cover letter]); the permanent public URL and archival DOI will be provided on acceptance. Every number in this manuscript is regenerated by a single script from archived out-of-fold predictions, requiring no imaging data and no GPU.

Imaging is redistributed by The Cancer Imaging Archive under its own terms and is not included in the repository. NSCLC-Radiogenomics is CC BY 3.0; NSCLC-Radiomics is CC BY-NC 3.0. Figure 6 shows NSCLC-Radiogenomics images only, consistent with that licence.

Figures



Figure 1: Participant flow. Counts and exclusion reasons are generated from the pipeline rather than transcribed, and reconcile exactly ($211 - 114 = 97$). The pre-specified primary analysis set is segmented adenocarcinoma.

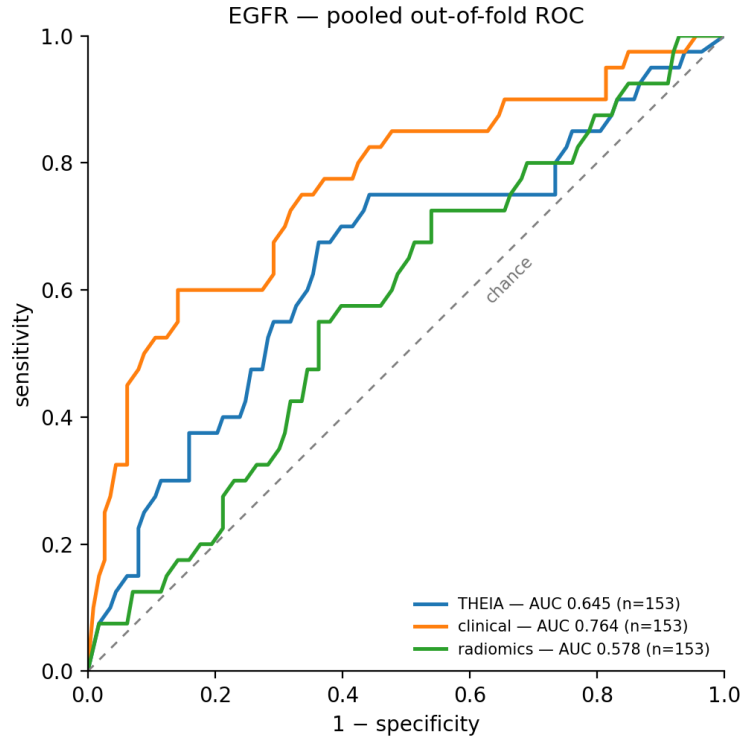


Figure 2: Pooled out-of-fold receiver operating characteristic curves by arm, on identical patients and folds. The chance line is drawn and labelled; each arm's n is printed, because arms do not all cover the same patients.

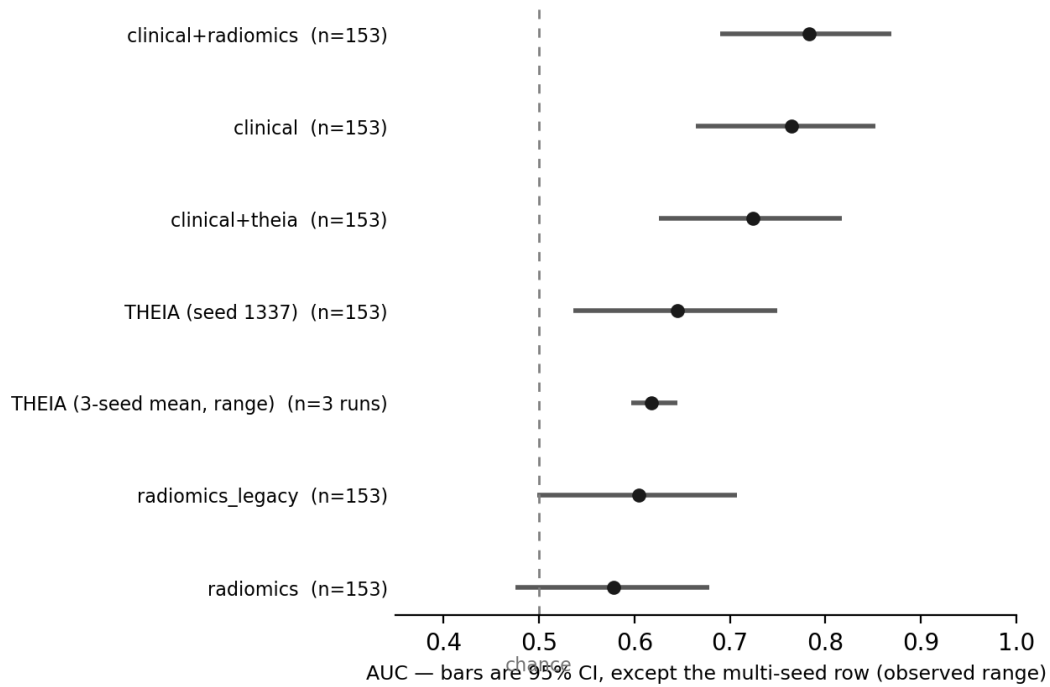


Figure 3: Every arm with its confidence interval. Overlapping intervals can still differ significantly when paired, which is why the paired deltas are reported alongside.

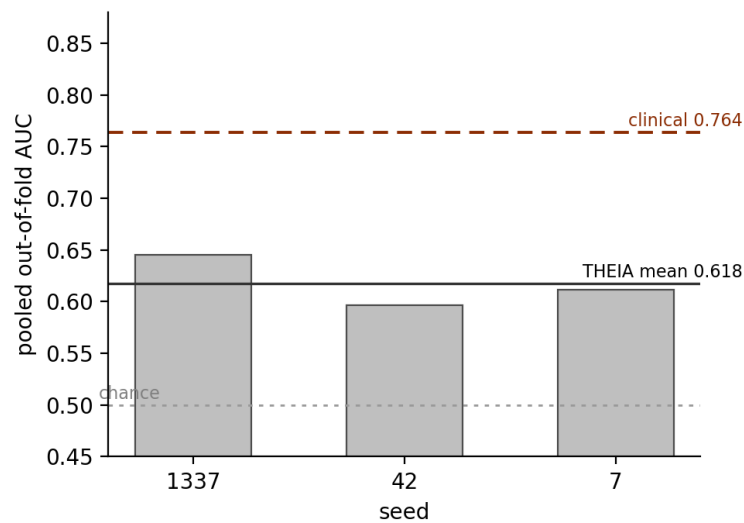


Figure 4: Per-seed AUC against the clinical baseline. The rerun spread is shown next to the effect being claimed; the gap to a five-variable clinical model exceeds it.

External validation: 420 held-out NSCLC-Radiomics patients (Maastricht, NL) — mass lift +0.244, beats shuffle 11/15

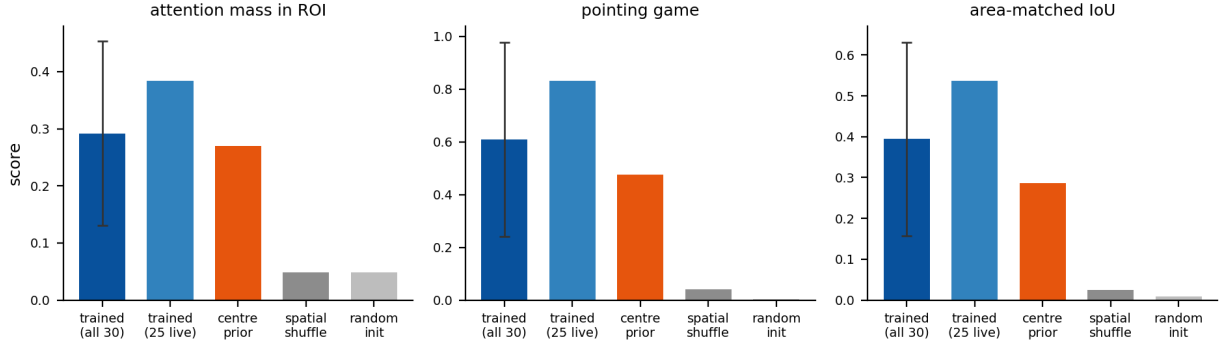


Figure 5: Primary endpoint: localisation on 420 held-out NSCLC-Radiomics patients, with all three controls in every panel. The centre prior is strong on attention mass by construction and weak on pointing accuracy and area-matched intersection over union, which are invariant to its width — so the trained model’s margin is read from the right-hand panels, not the left.

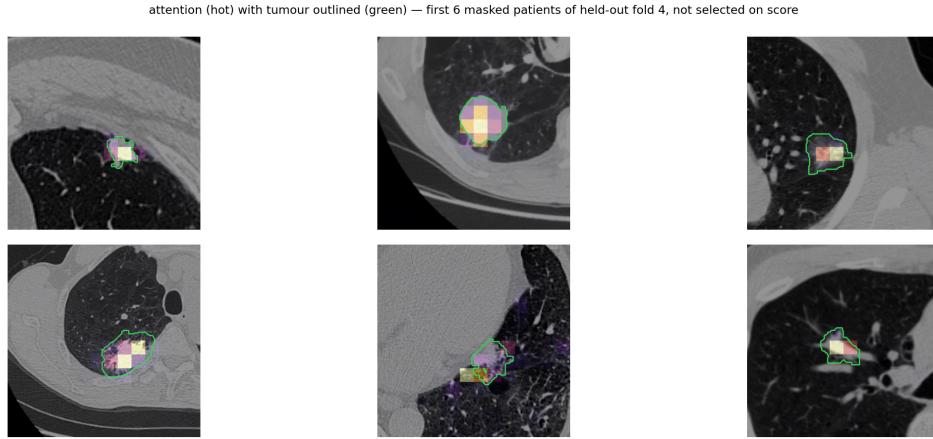


Figure 6: Attention maps with the tumour contour drawn from the reference segmentation. The first six segmented patients of a held-out fold in index order, *not* selected on score. Images from NSCLC-Radiogenomics (CC BY 3.0) [1].

Tables

Table 1: Cohort characteristics by *EGFR* status, restricted to the 153 patients with a known call. Never-smokers are 50% of mutants against 12% of wild-type; this is the textbook epidemiology and is most of what the clinical model knows.

Characteristic	<i>EGFR</i> mutant	Wild-type	All
<i>n</i>	40	113	153
Age (y), mean $\pm$ SD	—	—	69 $\pm$ 9
Age (y), median [IQR]	68 [63–76]	69 [64–75]	69 [64–76]
Pack-years, median [IQR]	15 [9–34]	42 [26–56]	40 [20–50]
Female	22 (55%)	33 (29%)	55 (36%)
Male	18 (45%)	80 (71%)	98 (64%)
Never-smoker	20 (50%)	13 (12%)	33 (22%)
Former smoker	18 (45%)	78 (69%)	96 (63%)
Current smoker	2 (5%)	22 (19%)	24 (16%)
Asian	9 (22%)	6 (5%)	15 (10%)
Caucasian	13 (32%)	77 (68%)	90 (59%)
Other / not recorded	18 (45%)	30 (27%)	48 (31%)
Adenocarcinoma	40 (100%)	93 (82%)	133 (87%)
Squamous cell carcinoma	0 (0%)	17 (15%)	17 (11%)
NSCLC not otherwise specified	0 (0%)	3 (3%)	3 (2%)

Note.—IQR = interquartile range, NSCLC = non-small cell lung cancer. Percentages are column percentages within each *EGFR* group. Age is summarised by both mean $\pm$ SD and median [IQR] because the distribution is mildly left-skewed; per-group means are omitted to avoid implying a comparison that was not tested.

Table 2: All arms and fusion topologies, on identical patients and folds. No combination of the imaging model with clinical variables exceeds the clinical model alone, and adding radiomics significantly harms it.

Arm	<i>EGFR</i> AUC	Δ vs clinical	<i>P</i> value
<i>Single arms</i>			
Smoking status alone	0.794	+0.030	—
Clinical (5 variables)	0.764	—	—
THEIA, 3 seeds	0.618 ± 0.025	−0.108	—
+ peritumoral branch	0.675 ± 0.017	−0.089	—
Frozen-feature logistic probe	0.617 ± 0.052	−0.147	—
Radiomics	0.578	−0.186	—
<i>Fusion topologies</i>			
Soft voting (model + clinical)	0.748	−0.016	.59
Stacking (cross-fitted)	0.735	−0.029	.11
Soft voting (equal weights)	0.712	−0.052	.15
Soft voting (clinical + radiomics)	0.703	−0.061	.02
Hard voting	0.690	−0.074	.07
Soft voting (model + radiomics)	0.634	−0.130	.02
<i>Co-primary endpoint</i>			
Δ AUC (clinical + model) − clinical	−0.029 $\pm$ 0.015 , every seed’s CI includes zero		

Note.—AUC = area under the receiver operating characteristic curve, CI = confidence interval. Δ is against the clinical model. *P* values are from a paired bootstrap over patients (2000 resamples); two-sided *P* < .05 indicated significance. Values with $\pm$ are across-seed mean $\pm$ SD over three seeds.

Table 3: Primary endpoint, every external evaluation individually, on the same 420 held-out NSCLC-Radiomics patients. The margin over the pre-specified fold-rate threshold is one evaluation, so each is shown rather than summarised. Success and failure are separated by a wide gap on every metric: no evaluation falls between 0.000 and 0.757 on pointing accuracy.

Run	Fold	Mass	Shuffle	Lift	Pointing	IoU
A	0	0.375	0.048	+0.327	0.862	0.563
A	1	0.038	0.045	−0.007	0.000	0.001
A	2	0.394	0.048	+0.346	0.855	0.543
A	3	0.038	0.045	−0.007	0.000	0.000
A	4	0.043	0.048	−0.005	0.000	0.000
B	0	0.327	0.049	+0.278	0.833	0.538
B	1	0.316	0.048	+0.268	0.790	0.502
B	2	0.401	0.050	+0.351	0.893	0.570
B	3	0.411	0.051	+0.360	0.831	0.562
B	4	0.550	0.049	+0.501	0.802	0.525
C	0	0.345	0.049	+0.296	0.810	0.524
C	1	0.040	0.044	−0.005	0.000	0.016
C	2	0.306	0.049	+0.258	0.757	0.469
C	3	0.409	0.051	+0.359	0.829	0.550
C	4	0.384	0.048	+0.336	0.874	0.553
Trained, mean		0.292	0.048	+0.244	0.609	0.394
<i>Control: centre prior</i>		0.270	0.056	+0.214	0.476	0.287
<i>Control: untrained head</i>		0.049	0.049	−0.000	0.005	0.008

Note.—IoU = area-matched intersection over union. Runs A, B and C are three independently seeded training runs of five cross-validation folds each. Mass is the fraction of attention falling inside the reference tumour contour; Shuffle is the same values permuted across cells; Lift is their difference. Eleven of 15 evaluations exceed their shuffle, meeting the pre-specified threshold of 71.4% by one evaluation. All 11 successes exceed the centre prior on pointing accuracy.

References

- [1] Bakr S, Gevaert O, Echegaray S, et al. A radiogenomic dataset of non-small cell lung cancer. *Sci Data* 2018;5:180202. doi:10.1038/sdata.2018.202
- [2] Gevaert O, Echegaray S, Khuong A, et al. Predictive radiogenomics modeling of EGFR mutation status in lung cancer. *Sci Rep* 2017;7:41674. doi:10.1038/srep41674
- [3] Aerts HJWL, Wee L, Rios Velazquez E, et al. Data From NSCLC-Radiomics (version 4). The Cancer Imaging Archive, 2014. doi:10.7937/K9/TCIA.2015.PF0M9REI
- [4] Rodríguez Sánchez DI, Middelkoop J, Vanneste T, et al. Quality over quantity: biopsy-anchored CT radiogenomics models outperform all-lesion training in a multi-tumour cohort despite a smaller sample size. *Eur Radiol* 2026. doi:10.1007/s00330-026-12601-9
- [5] Huang L, Xu L, Wang X, et al. Prediction of EGFR mutations in lung adenocarcinoma via CT images. *Acad Radiol* 2025;32(8):4880–4892. doi:10.1016/j.acra.2025.04.029
- [6] Shang Y, Chen W, Li G, et al. Computed tomography-derived intratumoral and peritumoral radiomics in predicting EGFR mutation in lung adenocarcinoma. *Radiol Med* 2023;128(12):1483–1496. doi:10.1007/s11547-023-01722-6
- [7] Yamazaki M, Yagi T, Tominaga M, et al. Role of intratumoral and peritumoral CT radiomics for the prediction of EGFR gene mutation in primary lung cancer. *Br J Radiol* 2022;95(1140):20220374. doi:10.1259/bjr.20220374
- [8] Wu J, Meng H, Zhou L, et al. Habitat radiomics and deep learning fusion nomogram to predict EGFR mutation status in stage I non-small cell lung cancer. *Sci Rep* 2024;14:15877. doi:10.1038/s41598-024-66751-1
- [9] Le MN, Nguyen TM, Ngo LL, et al. Explainable 1-mm peritumoral CT radiomics for EGFR mutation prediction in non-small cell lung cancer. *Cancer Control* 2026;33:10732748261455532. doi:10.1177/10732748261455532
- [10] Lai R, Sheng D, Geng Y, et al. The predictive value of ¹⁸F-FDG PET/CT habitat radiomics combined model in evaluating EGFR gene mutations in lung adenocarcinoma. *Front Med* 2026;13:1868229. doi:10.3389/fmed.2026.1868229
- [11] Maleki F, Ovens K, Gupta R, Reinhold C, Spatz A, Forghani R. Generalizability of machine learning models: quantitative evaluation of three methodological pitfalls. *Radiol Artif Intell* 2022;5(1):e220028. doi:10.1148/ryai.220028
- [12] Horvat N, Papanikolaou N, Koh DM. Radiomics beyond the hype: a critical evaluation toward oncologic clinical use. *Radiol Artif Intell* 2024;6(4):e230437. doi:10.1148/ryai.230437
- [13] Hanley JA, McNeil BJ. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology* 1982;143(1):29–36. doi:10.1148/radiology.143.1.7063747
- [14] Riley RD, Ensor J, Snell KIE, et al. Calculating the sample size required for developing a clinical prediction model. *BMJ* 2020;368:m441. doi:10.1136/bmj.m441
- [15] Mongan J, Moy L, Kahn CE Jr. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): a guide for authors and reviewers. *Radiol Artif Intell* 2020;2(2):e200029. doi:10.1148/ryai.2020200029